# GIS Driven Land Utilization Analysis for Coconut Genebanks

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Project Proposal Report

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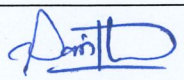
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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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# ABSTRACT

One of Sri Lanka's most economically significant plantation crops, coconuts, are essential to the nation's agricultural economy. To protect genetic diversity and assist with research, the Coconut Research Institute of Sri Lanka (CRISL) keeps a number of coconuts genebanks in various locations. However, it is challenging to perform integrated analysis and comprehend regional variations in coconut tree characteristics because the majority of current genebank records are kept in separate locations and analysed using manual or tabular methods. Researchers' capacity to carry out extensive data-driven analysis and spot regional trends in coconut performance is hampered by the lack of a spatially integrated system. While there are several agricultural information systems, the majority concentrate on data storage, crop yield prediction, or land suitability analysis; very few systems combine geospatial analytics with coconut genebank data to offer predictive and visual insights. This study suggests a GIS-based spatial intelligence framework for coconut genebank analysis to overcome this constraint. The system uses Geographic Information Systems (GIS) to perform spatial analysis and create heatmaps that show regional differences in coconut characteristics by integrating genebank data with geographic location information. To identify trends and forecast high-performing and low-performing areas, machine learning techniques are further utilized. The suggested approach makes use of tools like QGIS, PostGIS, Python-based data processing, and Scikit-learn-implemented machine learning models. The anticipated results include enhanced decision-support capabilities for agricultural researchers and plantation planners, better identification of regional performance trends, and improved spatial visualization of coconut genebank data.


**Keywords:** Coconut Genebank, Geographic Information Systems, Heatmap Analysis, Machine Learning, Spatial Agriculture.

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# LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| CRI | Coconut Research Institute |
| CRISL | Coconut Research Institute of Sri Lanka |
| GIS | Geographic Information System |
| GPS | Global Positioning System |
| ML | Machine Learning |
| QGIS | Quantum Geographic Information System |
| SDG | Sustainable Development Goal |
| DSR | Design Science Research |
| WBS | Work Breakdown Structure |
| SaaS | Software as a Service |
| KDE | Kernel Density Estimation |
| API | Application Programming Interface |

*Table 1. ABBREVIATIONS*

# 1. INTRODUCTION

## 1.1 Background and Context

Coconut is one of the most important plantation crops in Sri Lanka and plays a significant role in the country's agricultural economy. It makes a significant contribution to national agricultural productivity, employment prospects, and rural livelihoods. The Coconut Research Institute of Sri Lanka (CRISL) maintains multiple coconut genebanks spread throughout the nation's coconut-growing regions to protect coconut genetic diversity and facilitate long-term agricultural research. These genebanks hold important data about the morphological features, growth patterns, and performance indicators of coconut trees, which are crucial for future coconut breeding initiatives, research, and conservation.

Understanding how land areas are used is crucial for enhancing plantation planning, resource allocation, and long-term crop management in large agricultural settings like coconut genebanks. Plantation areas where coconut trees are preserved for study and genetic conservation are part of the Coconut Research Institute of Sri Lanka's (CRI) coconut genebanks. These sites include both planted areas and vacant yard spaces that may be used for upcoming experiments and plantings.

However, conventional methods like manual observations or tabular records are frequently used to evaluate land utilization within these genebank environments. A clear spatial understanding of how land is distributed and used throughout the entire genebank is not provided by these methods, despite the fact that they offer basic information about plantation areas. Researchers find it challenging to decipher spatial patterns of land use when plantation boundaries, tree locations, and available garden spaces are only recorded as text or numbers.

Geographic Information Systems (GIS) provide a powerful tool for managing agricultural data. GIS combines geographic coordinates with information on plantations, which helps in visualizing the extent of the land through digital maps. This helps in analyzing the extent of the land that has been used for different purposes by performing a spatial analysis. This visualization technique helps in identifying the extent of the land that has been used by creating a heatmap. This helps in identifying the extent of the land that has been used or is available for further usage.

It is in light of these factors that this study proposes the development of a GIS-based spatial analysis system for assessing land utilization in coconut gene bank areas. This system will incorporate different aspects of spatial data, such as plantation boundary and tree spatial data, for analysis and creation of a heatmap for visualizing land usage intensity. This will, in turn, help in a better spatial visualization of land utilization and availability for more efficient land resource management in coconut plantations.

## 1.2 Research Problem

The land utilization in the coconut genebank is at present being assessed by conventional record-based methods that do not possess spatial visualization capabilities. The plantation boundaries and tree locations are being maintained in table format. This makes it difficult to interpret the land utilization distribution in the plantation.

It is difficult to identify the land utilization and available land areas in the plantation without spatial analysis capabilities. The manual interpretation of plantation records is a time-consuming process and does not provide accurate insights for effective land management in large areas.

The majority of the agricultural information systems emphasize data storage rather than spatial analysis and visualization [6], [7]. The use of techniques such as heatmap visualization for effective land utilization intensity in different locations is rare in land management in the genebank.

Therefore, there is a need for a GIS-based spatial analysis framework that incorporates spatial data and heatmap visualization to evaluate land utilization patterns. This would enable researchers to easily identify utilized and available land, thus facilitating effective plantation planning and management decisions.

## 1.3 Literature Review

Recent developments in geospatial technology and data analysis have enhanced agricultural monitoring and decision-making processes significantly. Geographic Information Systems (GIS) are now a major means of analyzing spatial agricultural data, helping agriculturalists visualize agricultural data, environmental data, and agricultural performance in specific regions of interest [8], [9]. GIS helps in analyzing agricultural data based on specific locations.

There are various research studies that have been conducted to examine the application of GIS technologies in agricultural suitability analysis. For example, Sawant et al. utilized GIS-based multi-criteria analysis to identify suitable areas for the plantation of coconuts. The authors incorporated environmental factors such as soil type, rainfall, and temperature in the analysis [1]. The research demonstrated the potential of spatial analysis in agricultural planning. Additionally, studies such as those by Rahim et al. highlighted the integration of GIS and remote sensing for land suitability assessment [2].

Other studies have also examined the integration of remote sensing and geospatial data for agricultural analysis. Rahim et al. presented a GIS and remote sensing framework for land suitability analysis by integrating environmental data with geospatial data for better decision-making in land management [2]. These studies have shown the importance of geospatial data in agricultural analysis; however, most studies have focused on environmental suitability rather than evaluating crop performance based on biological or genetic data.

Machine learning methods have also started being used in the agricultural sector for pattern recognition and prediction. In a study by Chen et al., the authors showed that machine learning methods could be used for predicting crop yields by integrating different data sources like climate and satellite data [3]. In another study, geospatial machine learning techniques were used for agricultural land suitability analysis [4]. Furthermore, machine learning has also been applied for coconut disease detection using image processing techniques [5].

Apart from these research works, platforms like Genesys and GRIN Global are available for centralized database management for plant genetic resources collected from different genebanks across the world [6], [7]. Yet these platforms are more focused on centralized database management rather than spatial analysis and prediction.

Despite these developments, limited research is available on the integration of coconut genebank data with GIS-based spatial visualization and prediction analysis. Most platforms are available for plant genetic resource management in a tabular format without heatmap-based spatial analysis. Hence, a platform for integrated genebank data, GIS analysis, heatmap visualization, and machine learning is needed for regional coconut-related research and decision-making.

| Study | Method / Technology | Application | Key Contribution | Limitations |
|---|---|---|---|---|
| Sawant et al. (2023) | GIS-based Multi-Criteria Decision Analysis | Land suitability assessment for coconut cultivation | Demonstrated how GIS can evaluate suitable coconut growing regions using environmental parameters | Does not integrate coconut genetic resource data or genebank datasets |
| Rahim et al. (2024) | GIS + Remote sensing | Agricultural land suitability planning | Integrated environmental data with GIS for crop suitability analysis | Does not include machine learning or heatmap visualization for crop performance |
| Chen et al. (2021) | Machine learning with satellite and climate data | Crop yield prediction | Showed that combining satellite imagery with environmental data improves agricultural prediction accuracy | Study focuses on large-scale crops such as maize, not coconut genebank analysis |
| Sathiyamurthi et al. (2024) | Geospatial ML + remote sensing | Agricultural land suitability modeling | Demonstrated integration of geospatial datasets and ML algorithms for land suitability analysis | Focused on land suitability rather than regional crop performance prediction |
| Genesys Genebank System | Global genetic resource information system | Genebank database management | Provides centralized access to plant genetic resource datasets worldwide | Focus on data storage rather than spatial analytics or predictive modeling |
| • Proposed Research | GIS + Heatmap Visualization + Machine Learning Prediction | Coconut genebank spatial intelligence system | Integrates genebank data from multiple regions into a centralized spatial database, generates heatmaps to visualize regional coconut performance, and applies ML models to predict high- and low-performing regions | Depends on availability and quality of genebank data |

*Table 2 Comparison of Existing Approaches vs Proposed Method*

## 1.4 Research Gap

Despite the popularity and use of Geographic Information Systems (GIS) and spatial analytics in the field of agriculture, the use of these tools in analyzing land utilization in the context of coconut genebanks remains limited. Most studies conducted in this field have focused on crop yield prediction, land suitability analysis, and plantation mapping based on environmental factors such as soil, rainfall, and temperature [1], [2], [4]. Though these studies can help identify suitable regions for planting or predict crop performance, they do not specifically address the need for analyzing land utilization.

Land utilization assessments are commonly carried out in various agricultural research environments, including coconut genebanks, through manual observations or simple tabular recording systems. These conventional approaches do not allow for a clear visualization of the land area that is currently utilized by coconut trees and that is still available for future planting. The visualization of land area is not possible without proper spatial visualization tools, making it difficult to identify underutilized or available land.

Another limitation that has been observed is that there are no intuitive visual tools that help researchers quickly interpret patterns of land utilization over large areas. Most agricultural data is presented in numerical or textual formats that require a high degree of manual interpretation. This is where visualization techniques like GIS-based heatmaps can be used to effectively represent the intensity of land utilization [10].

Despite the increasing use of geospatial technology in agriculture, limited research exists on GIS-based spatial analysis specifically for evaluating land utilization. Current systems are not using heatmap visualization techniques to represent both utilized land and available yard space.

There is a clear need for a spatial analysis system that integrates GIS and heatmap visualization to evaluate land utilization within coconut genebanks. The proposed research aims to address this gap by applying GIS-based spatial analysis techniques to plantation data, enabling the creation of heatmap visualizations for improved understanding and decision-making in land management.

# 2. OBJECTIVE

## 2.1 Main Objective

Development of a GIS-based spatial analysis system capable of generating heatmaps to evaluate land utilization within coconut genebanks, enabling identification of currently utilized areas and available yard space to support efficient plantation planning and management.

## 2.2 Specific Objectives

- Collection of spatial data pertaining to coconut genebank land areas, including planted land and available yard spaces.
- Geographic Information System (GIS) application for visualizing coconut plantation land areas.
- Development of heatmap visualization based on land utilization intensity in various parts of the coconut genebank.
- Identification of land areas that are not utilized or are available for possible future coconut planting activities.
- Integration of spatial analysis methods for comparing utilized land areas with available land areas for effective land management decisions.
- Development of an intuitive spatial visualization system that allows users to easily understand land utilization patterns for effective decision-making in coconut plantation activities.

# 3. METHODOLOGY

The methodology for this study is based on Design Science Research (DSR), which emphasizes a design and development approach in finding a solution for a given research problem, particularly in the field of technology. The purpose of this part of the methodology is to design a GIS-based spatial analysis system that will assess land utilization within coconut genebanks using heatmap visualization techniques. This system will assess how much land within a coconut genebank is already being utilized for planting purposes and how much yard space is available for coconut planting in the future. This system will be able to give a better understanding of land utilization within coconut genebanks by using Geographic Information Systems (GIS).

The methodology is comprised of various stages like data collection for space, preprocessing of geospatial data, analysis of space, heatmap generation, and system implementation. During the data collection stage, data regarding the spatial information of the coconut gene bank areas will be collected with the aid of researchers and officers of the Coconut Research Institute of Sri Lanka. The data to be collected will include the geographical coordinates of the areas, boundaries of the plantation, and the data regarding the areas already occupied by coconut trees or available yard spaces. Ethical considerations will be made to ensure that all the data collected regarding space is only used for academic purposes.

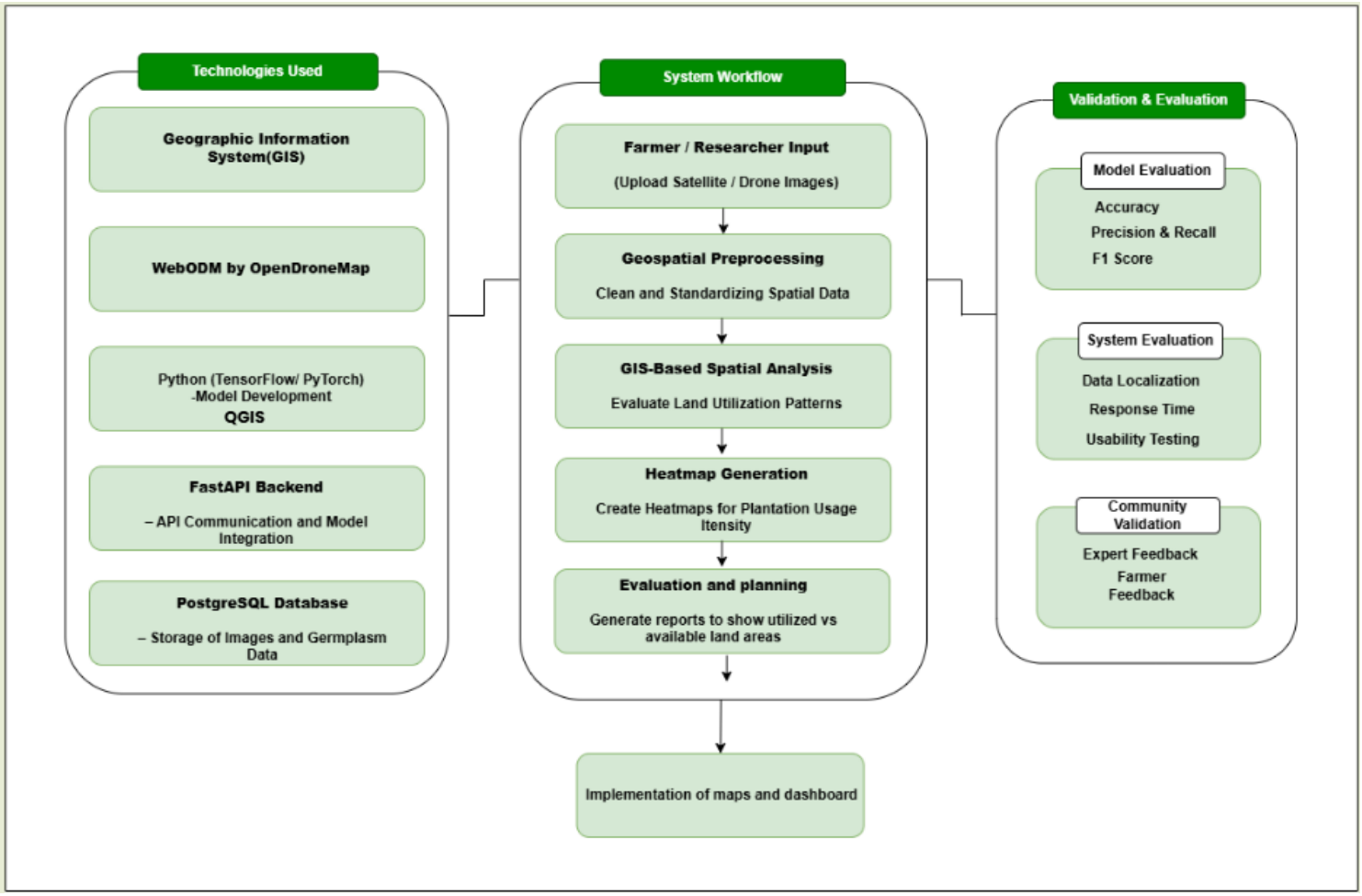
In the data preprocessing stage, the spatial data will be prepared for analysis. This will involve the application of various geospatial processing techniques to ensure consistency and accuracy of the data. The data will be prepared for analysis by the application of the above-mentioned techniques.

The GIS-based spatial analysis techniques will be used to analyze the patterns of land utilization within the genebank regions. The heatmap generation techniques will be used to determine the intensity of land usage. The heatmap generation techniques include kernel density estimation and intensity mapping. The heatmap will be used to identify the regions where the density of coconut planting is high. The heatmap will also be used to identify the regions where the land is not fully utilized for planting.

The proposed system will be implemented as a visualization tool that will enable the researcher to visualize the spatial maps as well as the heat maps generated for the genebank land utilization. The spatial analysis tool will be implemented using various GIS tools such as QGIS. The Python geospatial libraries will be utilized to create the heat maps. The geospatial libraries include GeoPandas and Folium.

Lastly, the proposed system will be evaluated based on the spatial accuracy validation. The heat maps generated will be compared with the actual records of the land utilization provided by the experts. The performance of the proposed system will be evaluated based on the spatial accuracy of the results generated as well as the ability of the system to visualize the utilized as well as the available land for the coconut genebank.

*Figure 1 Methodology*



# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system architecture shows the entire workflow of the GIS-based land utilization analysis system for coconut genebanks. The system architecture is based on a sequence of steps starting from the data collection to the end of the data visualization.

The system will start with the data collection from the coconut genebank locations. The data will include the plantation boundaries, the geographic coordinates of the coconut trees, and the available yard spaces. This data will form the primary input for the spatial analysis.

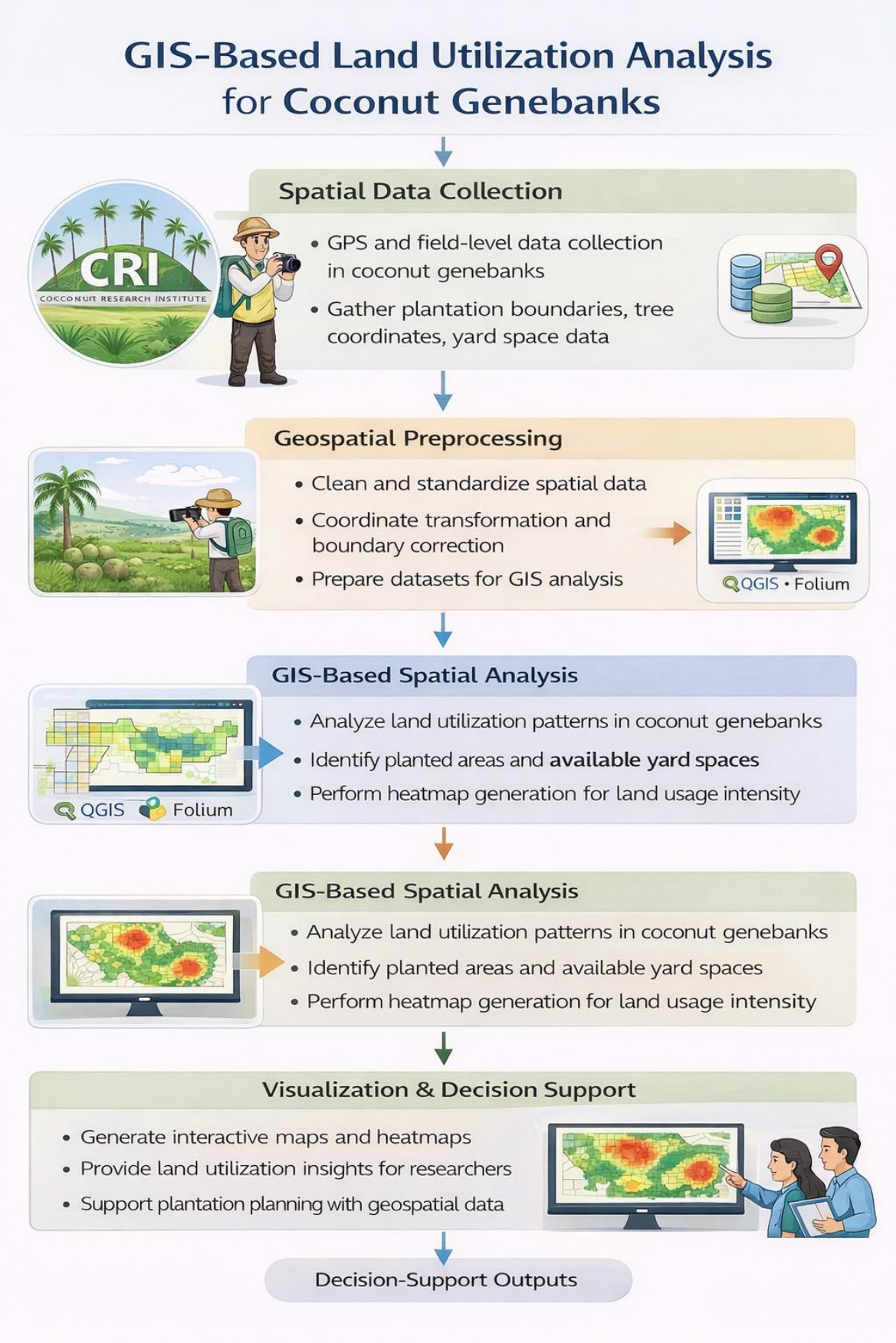
After the data is collected, the data will undergo the geospatial preprocessing stage. This stage will ensure that the data is properly processed for the spatial analysis. The geospatial preprocessing will ensure that the data is properly processed for the analysis.

After this preprocessing step, the spatial data is passed into the spatial analysis component. In this step, GIS technology is employed to compute land utilization patterns in the genebank regions. The system also identifies regions that are already occupied by coconut trees, as well as regions that are still available for future plantings.

After processing the spatial analysis, heatmap visualization is generated that represents land utilization patterns in the genebank regions. The heatmap visualization helps identify regions that are maximally utilized, as well as regions that are not utilized or are vacant.

Finally, the generated maps are presented through a visualization interface that is accessible by users of the system. This interface is useful for users who need to interpret maps and make decisions regarding plantation management strategies.

*Figure 2 High-Level System Architecture*



# 5. USER REQUIREMENTS

### Stakeholders

• Coconut Research Institute of Sri Lanka (CRI)

• Agricultural researchers

• Field officers / survey officers

• Plantation managers

### Functional Requirements

• The system should accept spatial data related to coconut genebank areas, including plantation boundaries and yard spaces.

• The system should allow input of GPS coordinates representing coconut tree locations or plantation areas.

• The system should process spatial datasets and prepare them for GIS analysis.

• The system should calculate the total land area currently utilized for coconut planting.

• The system should identify and calculate the remaining yard space available for new coconut plants.

• The system should generate GIS-based heatmaps representing land utilization intensity.

• The system should visually highlight highly utilized areas and underutilized areas within the genebank.

• The system should display spatial maps and heatmaps for interpretation by researchers and plantation managers.

### Non-Functional Requirements

• High spatial accuracy in land utilization analysis

• Fast processing time for GIS analysis and heatmap generation

• Scalability to support analysis of multiple genebank locations and larger spatial datasets

• User-friendly interface for viewing spatial maps and heatmap visualizations

# 6. COMMERCIALIZATION PLAN

The GIS-based land utilization analysis system for coconut genebank that has been proposed can be carried out by the Coconut Research Institute (CRI) in Sri Lanka and other relevant organizations that handle the land utilization for coconut genebanks and plantation research. The main users of the GIS-based land utilization analysis system for the coconut genebank will be the CRI staff and other relevant officials who handle the land utilization and future planting activities in the genebank.

The system offers a great value proposition since it is introducing a GIS-driven spatial analysis system that assesses land utilization while also showing the density of planting via a heatmap. The system can also analyze the spatial data for the genebank locations to identify the areas already utilized for coconut planting or those with available yard space for future planting. The proposed system is better compared to the traditional way of land assessment since it is more accurate, efficient, and can interpret the land usage patterns via a heatmap.

The platform can be utilized as a spatial visualization tool on the web. This will allow authorized users from various CRI regional genebank locations to view spatial maps and results of heatmap analysis. In terms of commercialization, the platform can be marketed to agricultural research institutions, plantation management companies, as well as government agencies that handle agricultural operations.

The key cost components related to the system are system development cost, cost of processing GIS data, cost of implementing spatial analysis, cloud hosting infrastructure cost, and system maintenance cost. The cost of system development can be kept relatively moderate since many geospatial technologies like QGIS and Python-based GIS libraries are open-source. The pricing models can be based on annual institutional subscription or license agreements based on the scale of implementation.

The proposed system has various competitive advantages. The system will incorporate a GIS-based spatial analysis tool that is specially designed for the context of coconut genebank land utilization. This will enable the easy visualization of the patterns of land utilization. Additionally, the system will allow for evidence-based plantation planning. This is because it will help to clearly identify the underutilization of the land in the context of the genebank. In terms of intellectual property rights, the software architecture developed will be subject to software copyright.

# 7. BUDGET AND JUSTIFICATION

| Resources | Estimated Price (LKR) |
|---|---|
| Transportation (visits to Coconut Research Institute for consultations and field observations) | 12,000.00 |
| Cloud server hosting and storage for spatial data and heatmap visualization platform | 8,000.00 |
| Internet and data usage (spatial data transfer, development, and testing) | 4,000.00 |
| Hardware / storage devices (external storage for spatial datasets and backups) | 6,000.00 |
| Stationery and documentation | 2,000.00 |
| Miscellaneous expenses | 3,000.00 |
| **Total** | **35,000.00** |

*Table 3 Budget Estimation*

## Budget Justification

### Transportation Costs

Transportation expenses are allocated for field visits to the Coconut Research Institute (CRI) and relevant genebank locations. These visits are necessary for understanding the plantation layout, consulting researchers, and verifying spatial information used in the GIS analysis.

### Cloud Hosting and Storage

Cloud infrastructure is required to host the GIS visualization platform and store spatial datasets. The cloud server enables remote access to maps and heatmaps, allowing researchers and planners to analyze land utilization results efficiently.

### Internet and Data Usage

Internet connectivity is necessary for transferring spatial datasets, accessing GIS resources, conducting development activities, and testing the visualization platform. Reliable connectivity ensures smooth integration of GIS tools and data processing.

**Hardware / Storage Devices**

External storage devices are required to store spatial datasets, map layers, analysis outputs, and backups. Since geospatial data files can be large, additional storage capacity is necessary to maintain research datasets safely.

**Stationery and Documentation**

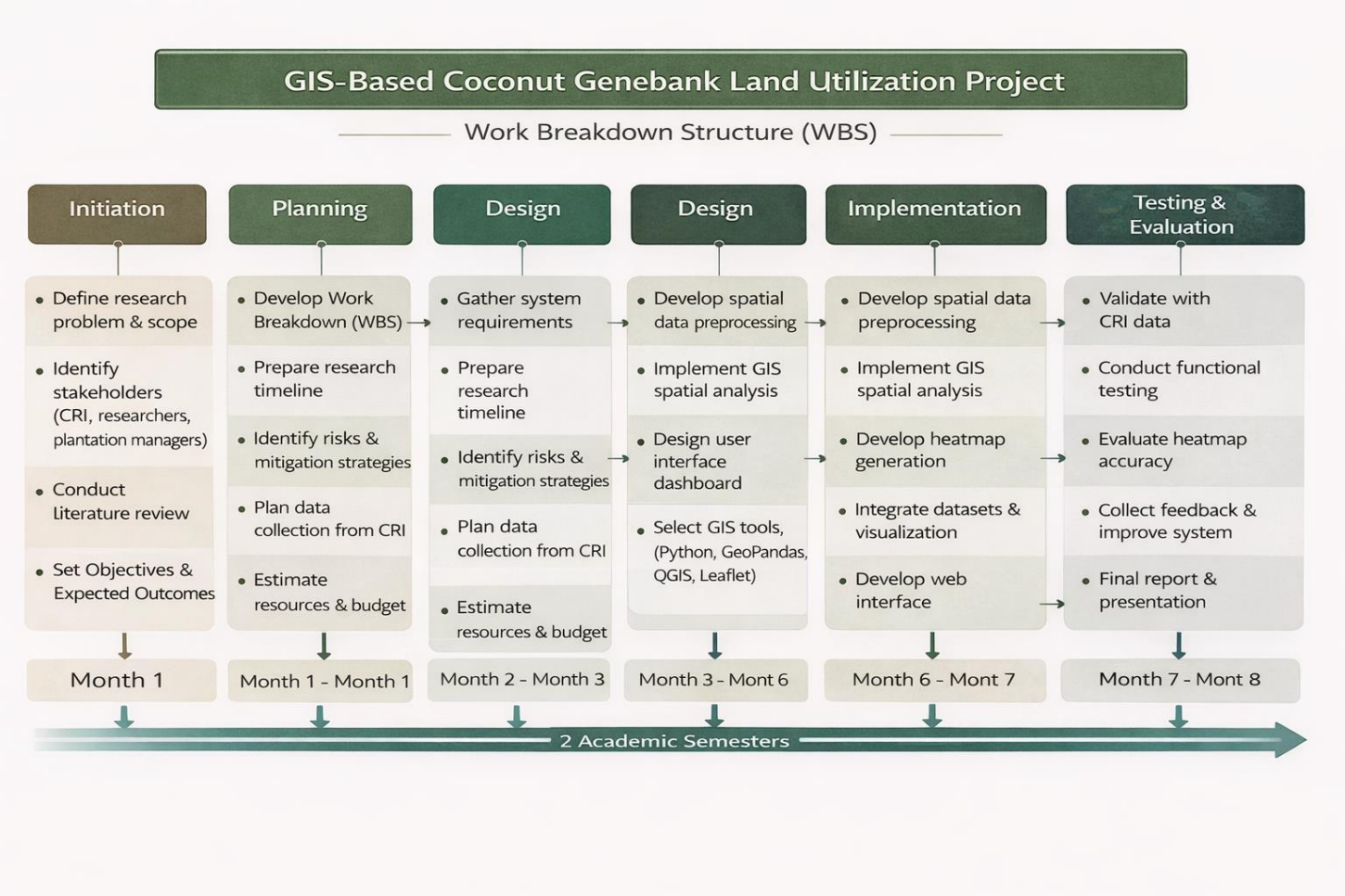
Stationery costs support documentation activities such as printing reports, preparing diagrams, and maintaining research records throughout the project lifecycle.

**Miscellaneous Expenses**

Miscellaneous expenses cover unforeseen costs that may arise during the research process, such as additional software utilities, small hardware accessories, or additional data collection needs.

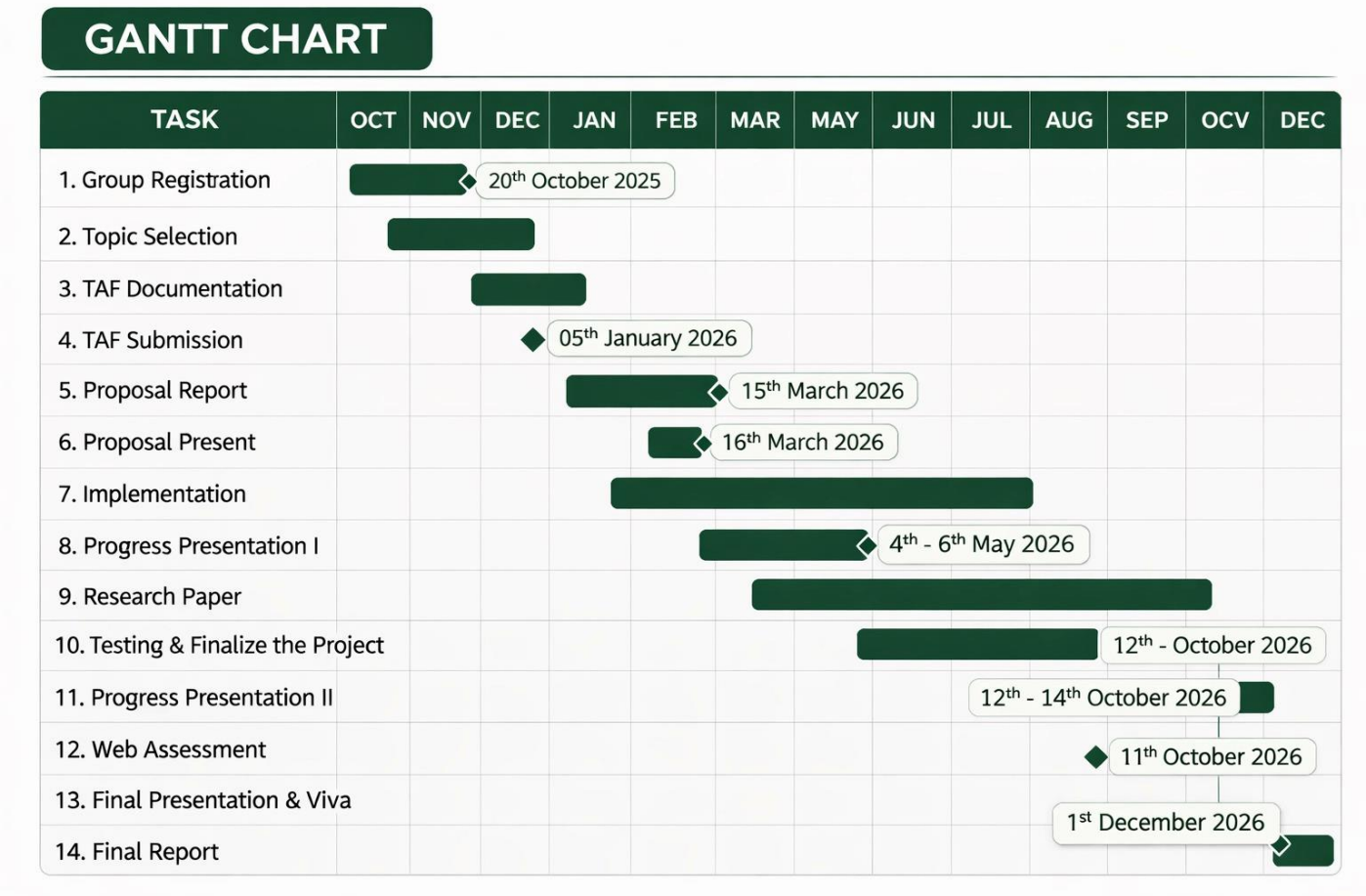
# 8. WORK BREAKDOWN STRUCTURE (WBS)

*Figure 3 Work Breakdown Structure*



# 9. GANTT CHART

*Figure 4 Gantt Chart*



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